EagleEye passenger model: training report

Male / female / child classifier, person detector, face model and seat-state model, from first run to the deployed car profile

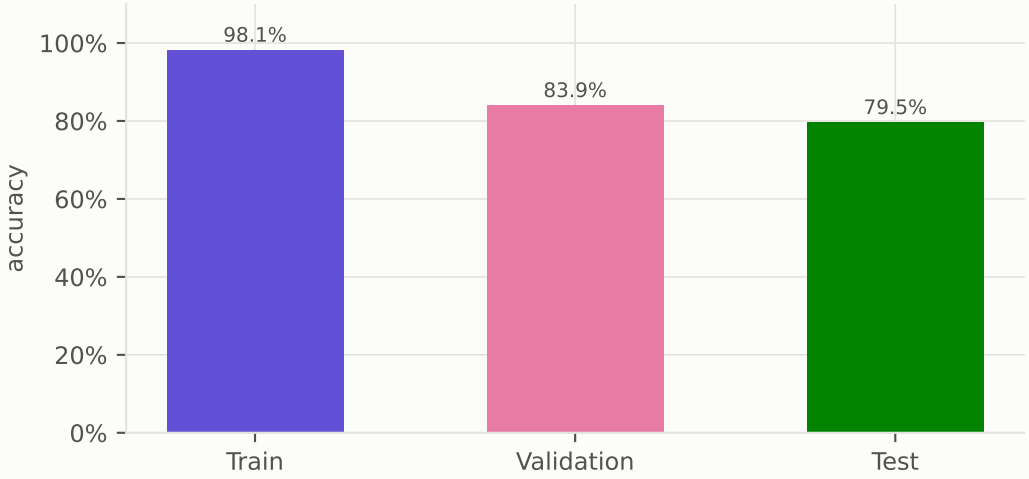
98%
Train accuracy (2,057 crops)
deployed offsets 0 / 1.5 / 1.0

84%
Validation accuracy (609 crops)
102 distinct person groups

80%
Test accuracy (611 crops)
only 14 groups, about 20 people

0.10
Test recall spread (lower = even)
was 0.38 before calibration

Accuracy by split: the generalisation gap



Recall per class (deployed model)



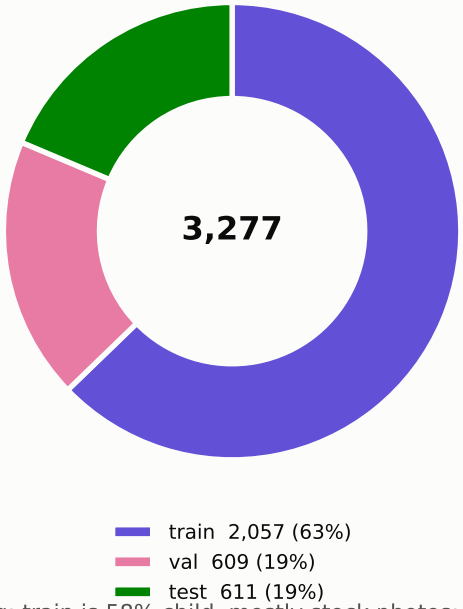
Model inventory

Person detector	YOLOv8n, COCO-pretrained, fine-tuned (v2); pipeline still uses plain COCO	test mAP50 0.835
Crop classifier	MobileNetV3-small, 112 px upper-body crop, 3 seeds, PA-100K pretrain	test acc 0.80
Face classifier	FairFace + UTKFace (bus profile); rejected for the car crop path	gender acc 0.898
Seat-state model	YOLOv8n occupied / unoccupied, bus busCabin data (v2b deployed)	test mAP50 0.804
Not trained	seatbelt, no-seatbelt, child-seat, 6-class cabin model	no weights

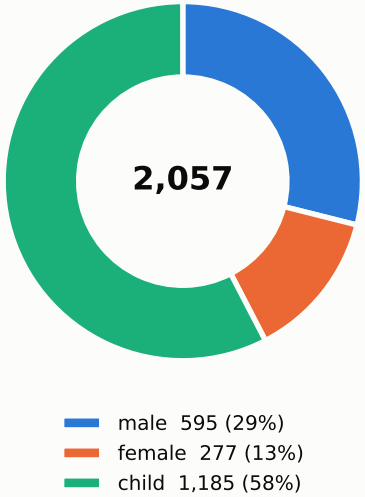
Training data: what the models learned from

crops_mfc_v4 (classifier) and person_det_v2 (detector). Counts are crops or images, not distinct people.

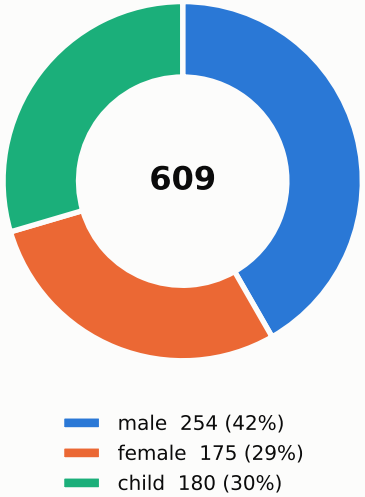
Classifier crops by split



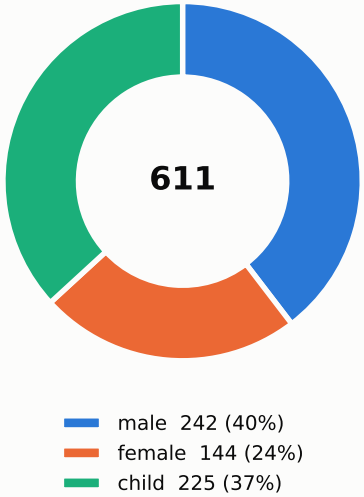
train: class mix



val: class mix

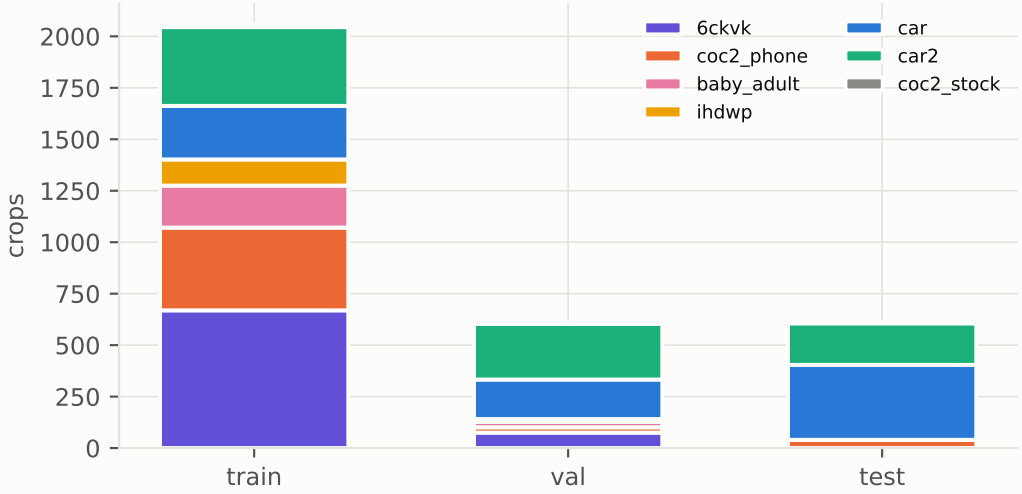


test: class mix

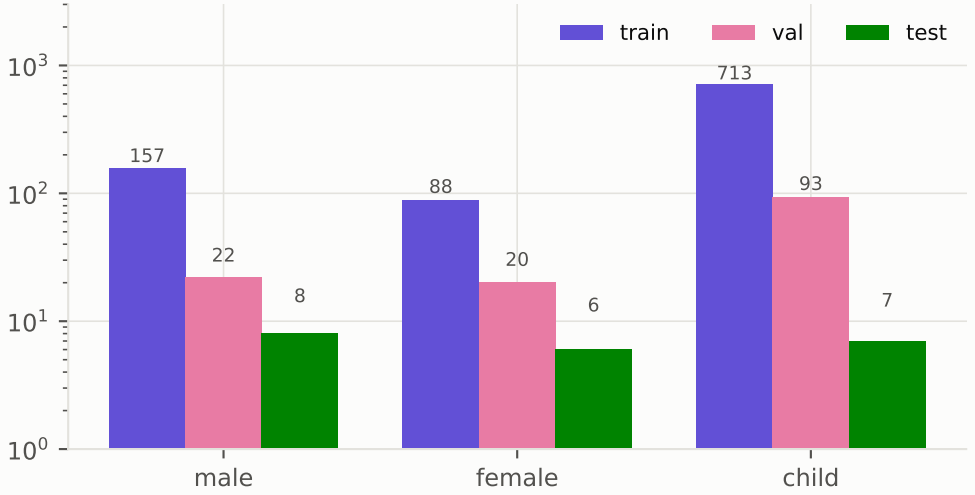


Reading: train is 58% child, mostly stock photos; test comes from 14 groups, one family video contributes 300 of 611 crops.

Crops by source and split



Distinct groups (source image / video) per class, log scale



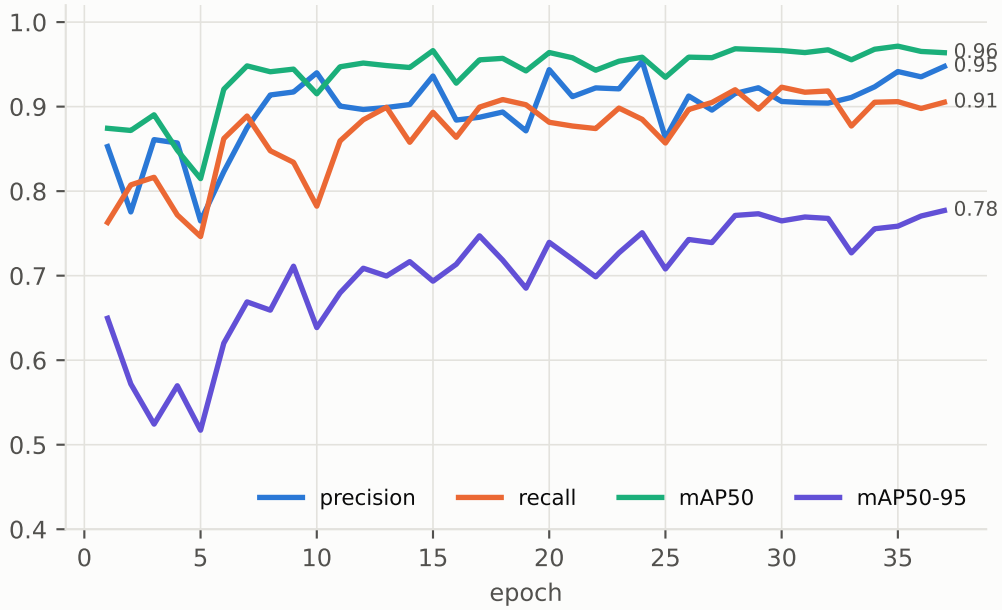
Person detector training: YOLOv8n fine-tune (person_det_v2)

37 epochs, 416 px, train 1,579 / val 355 / test 238 images. Per-epoch validation curves.

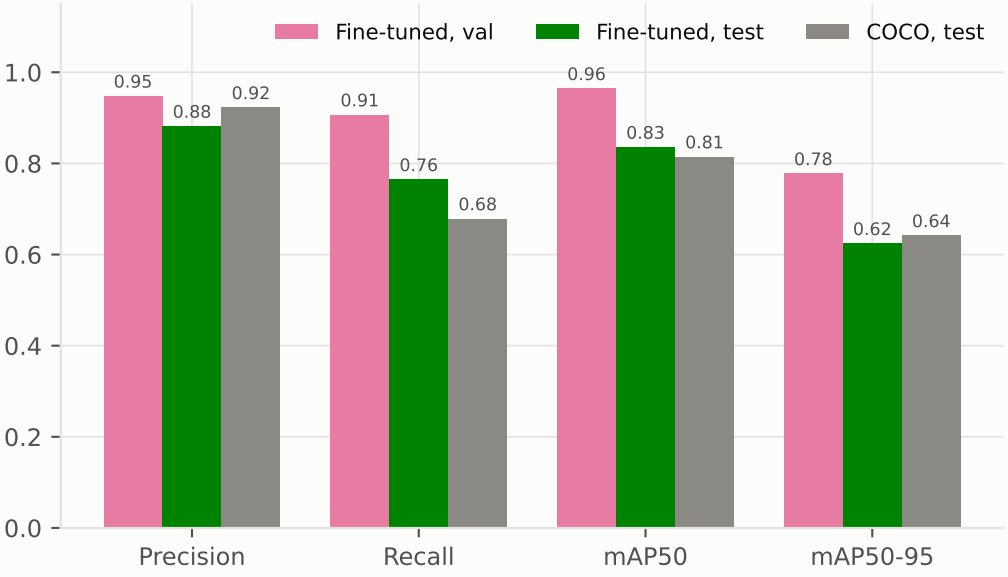
Loss per epoch



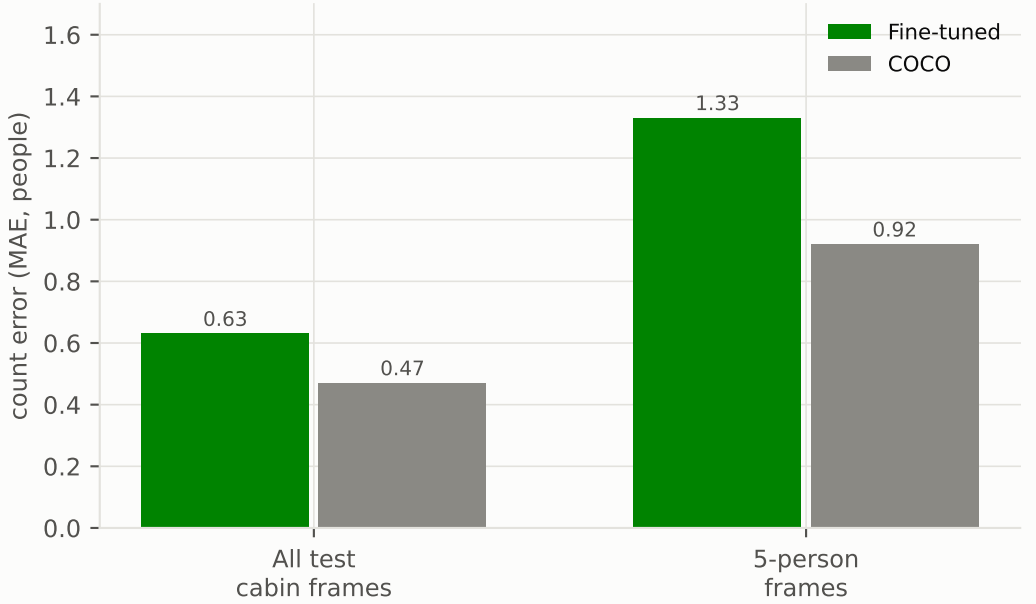
Validation precision, recall, mAP per epoch



Fine-tuned vs plain COCO YOLOv8n



Head-count error on test (lower is better)



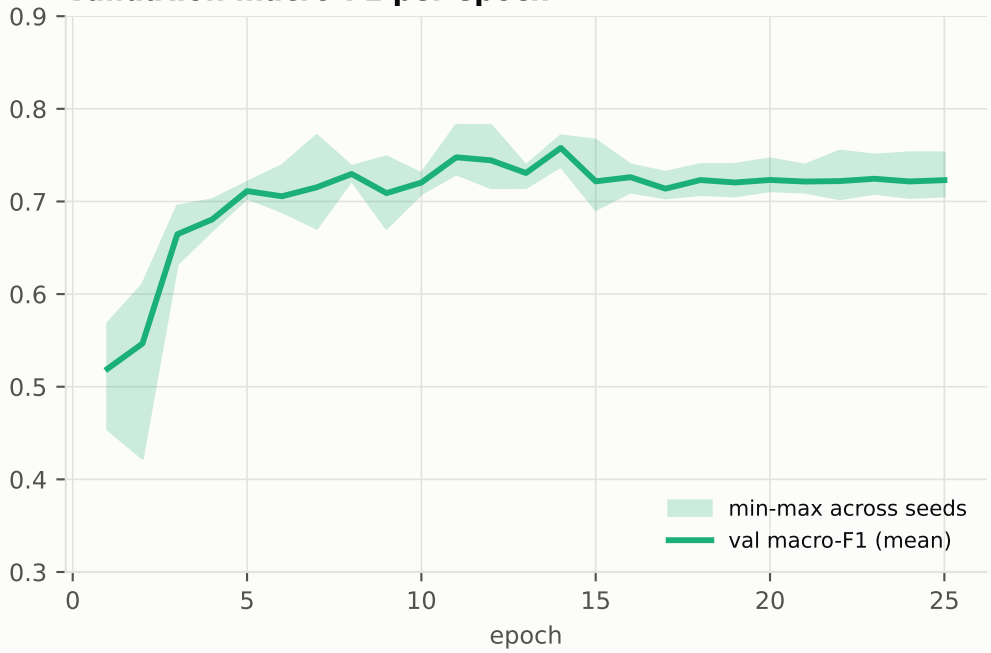
Classifier training curves (v4, mean of 3 seeds)

MobileNetV3-small, 25 epochs after PA-100K pretraining. Per-epoch accuracy was not logged; validation macro-F1 is shown instead.

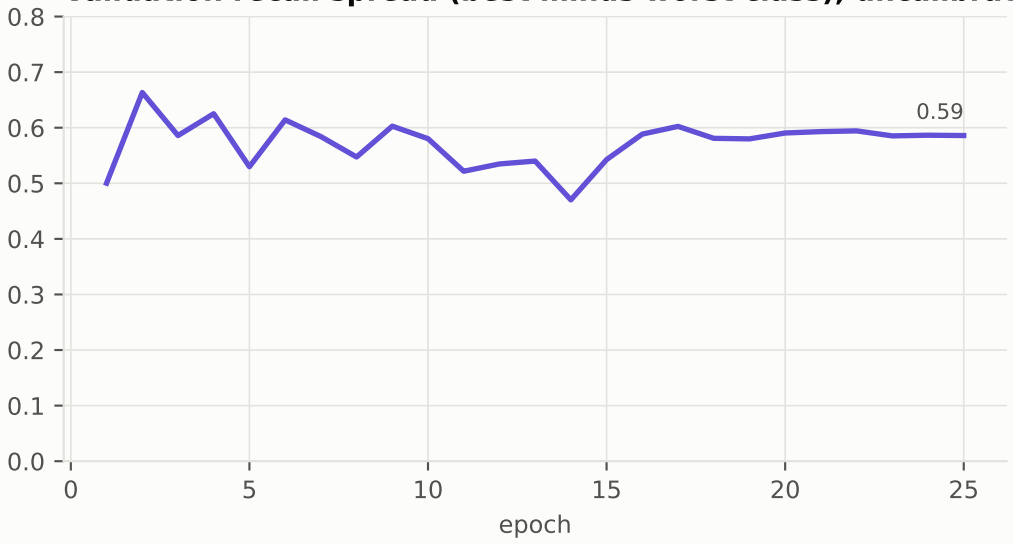
Loss: train keeps falling, validation turns up (overfitting)



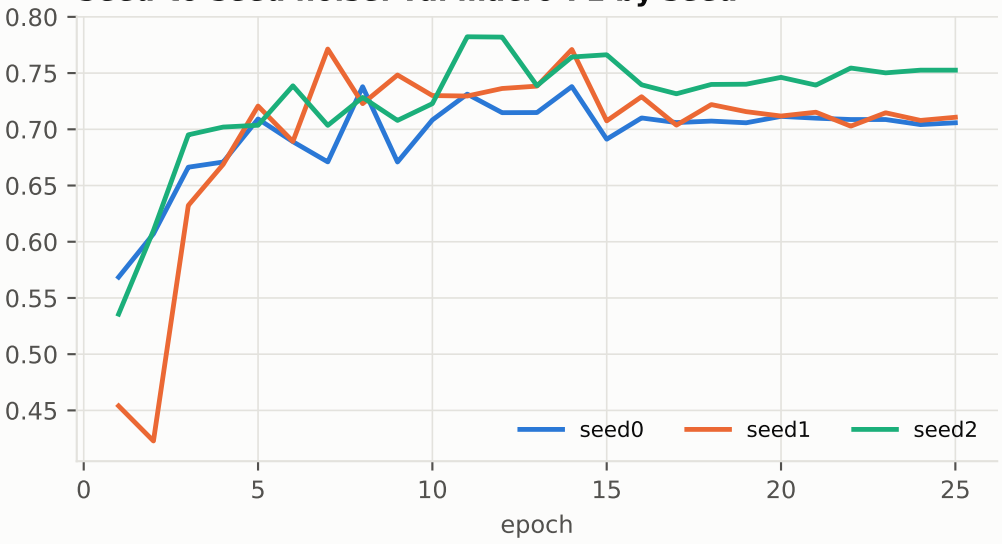
Validation macro-F1 per epoch



Validation recall spread (best minus worst class), uncalibrated



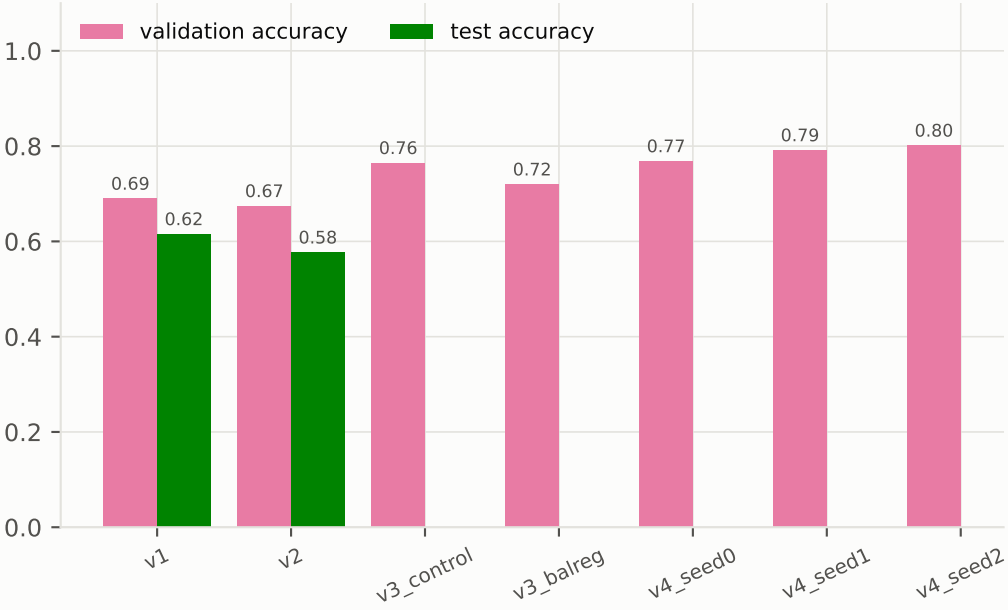
Seed-to-seed noise: val macro-F1 by seed



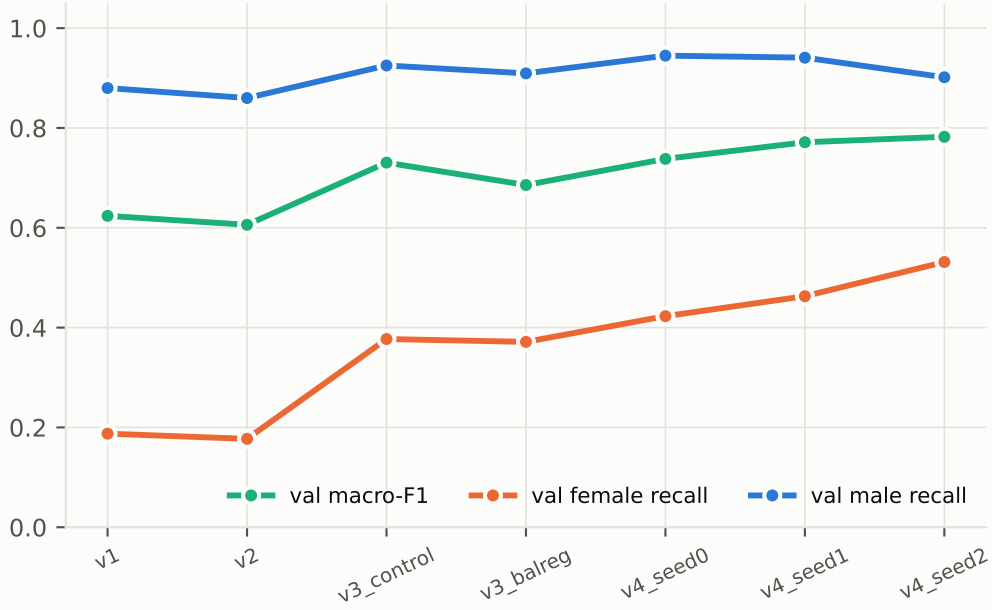
Classifier version history: what each training round gained

Validation and test numbers as stored in each run's metrics.json. Datasets changed between rounds, so versions are not strictly like for like.

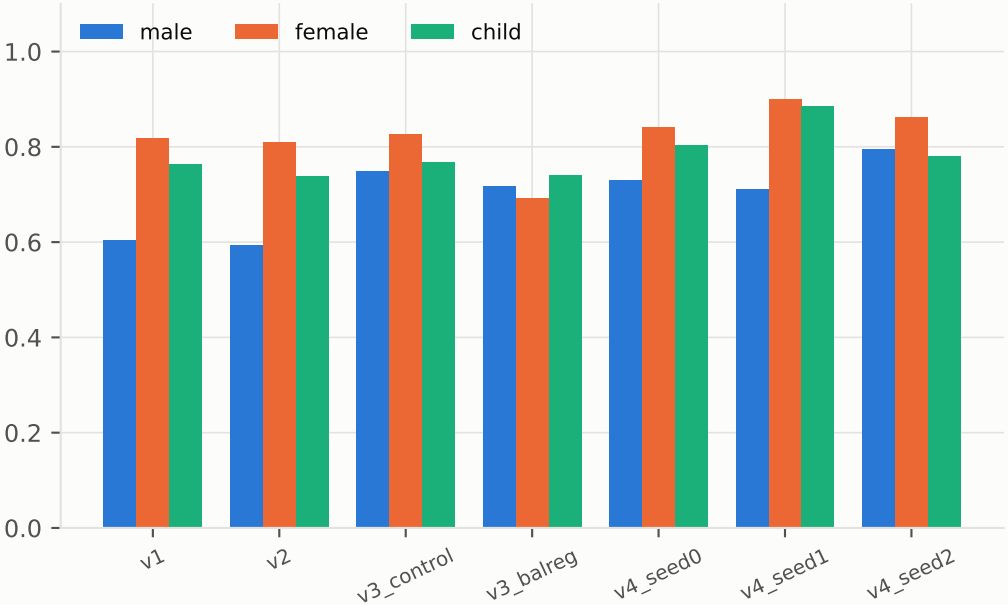
Accuracy by version (test scored only for v1, v2)



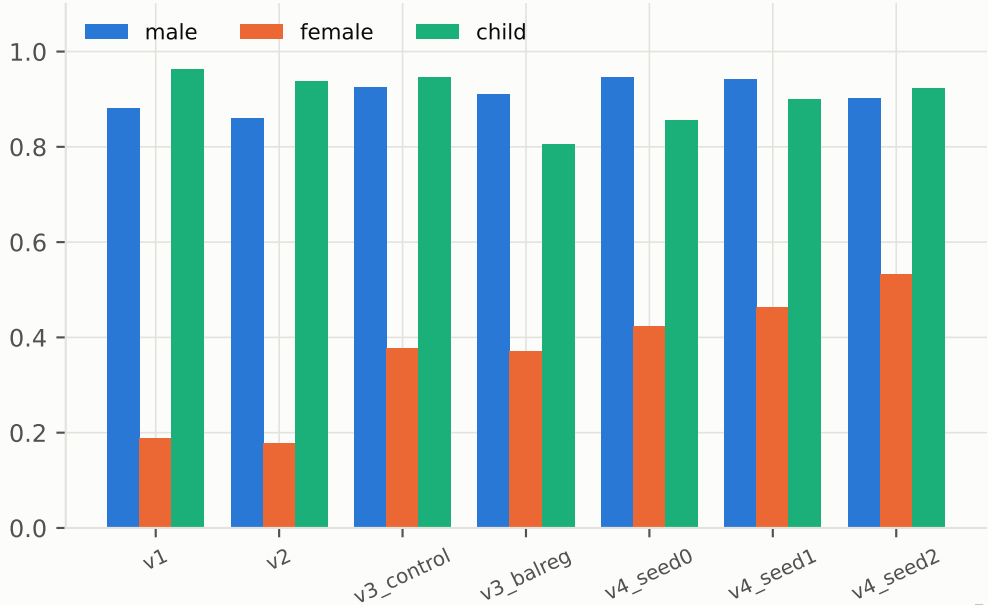
Uncalibrated validation: macro-F1 and recall by version



Validation precision per class

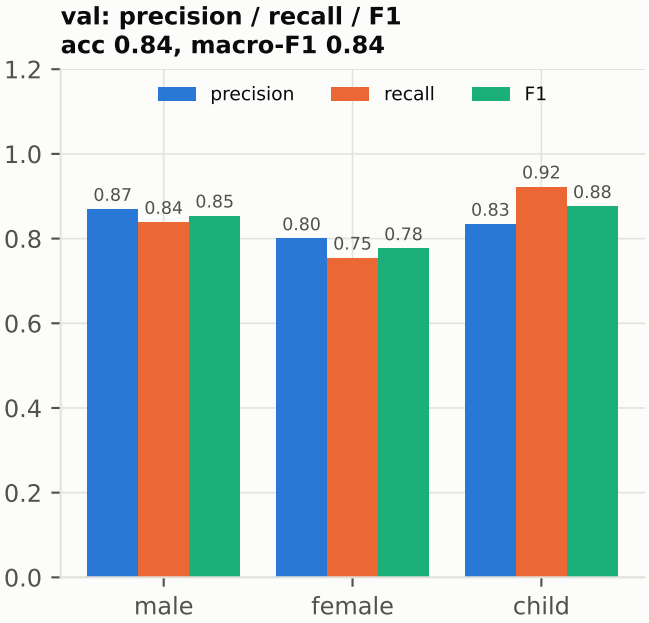


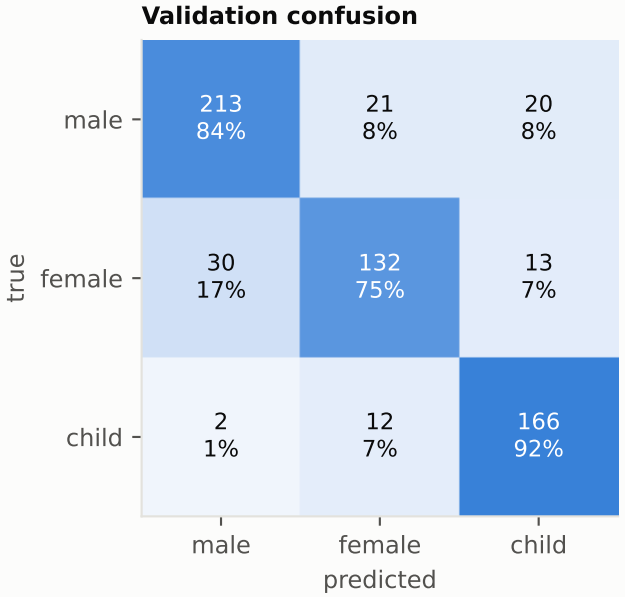
Validation recall per class

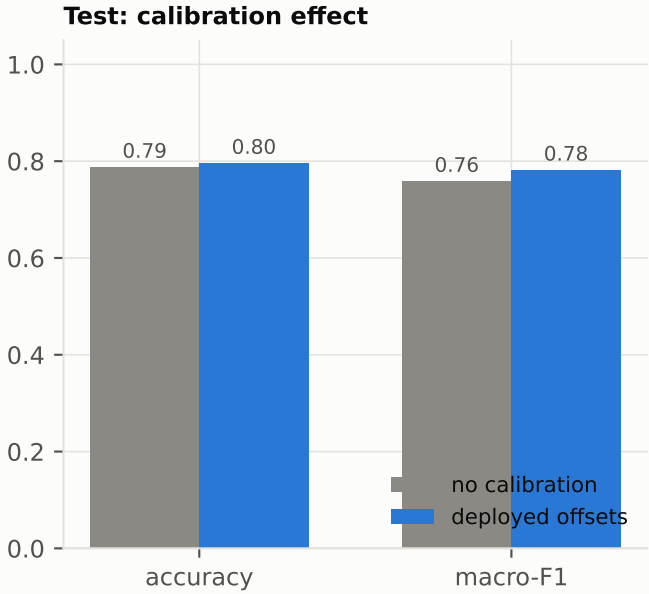


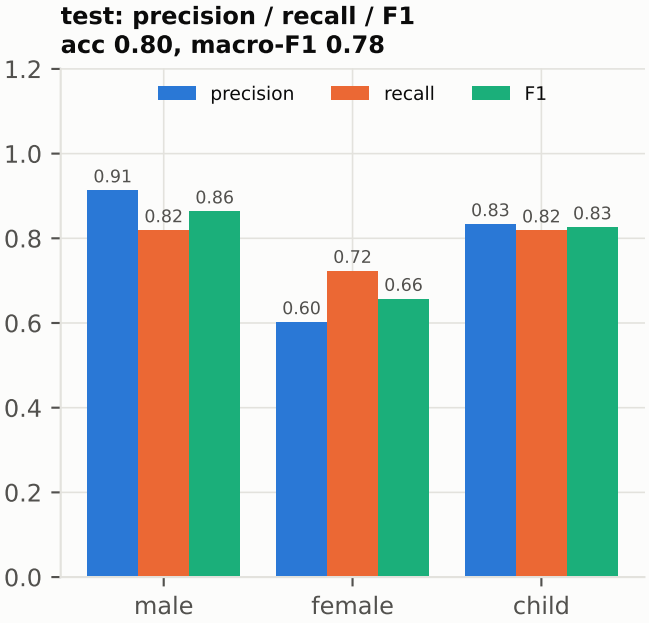
Deployed classifier: precision, recall, F1 and confusion

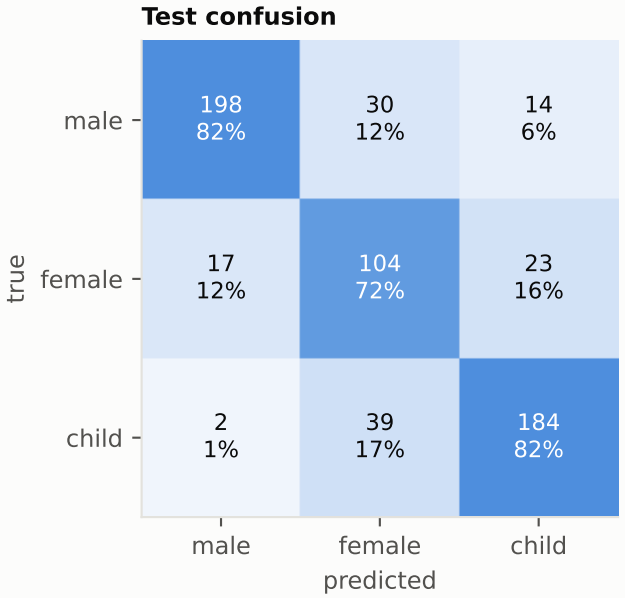
v4 three-seed ensemble with log offsets male 0 / female 1.5 / child 1.0 (the car profile in config.yaml). Recomputed on every saved crop.

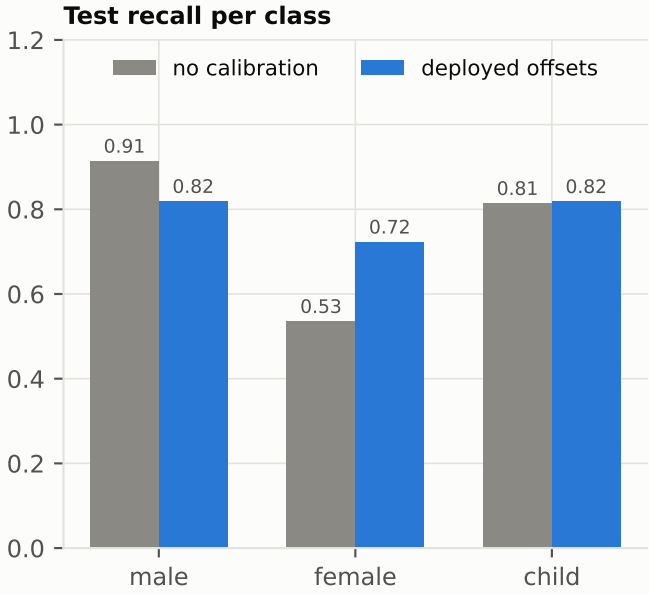








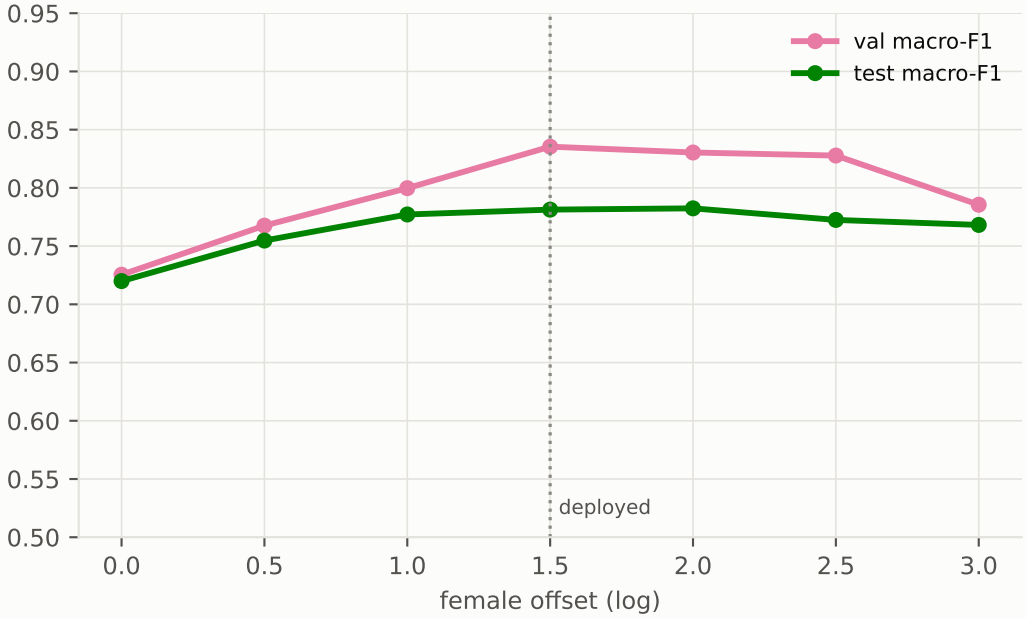




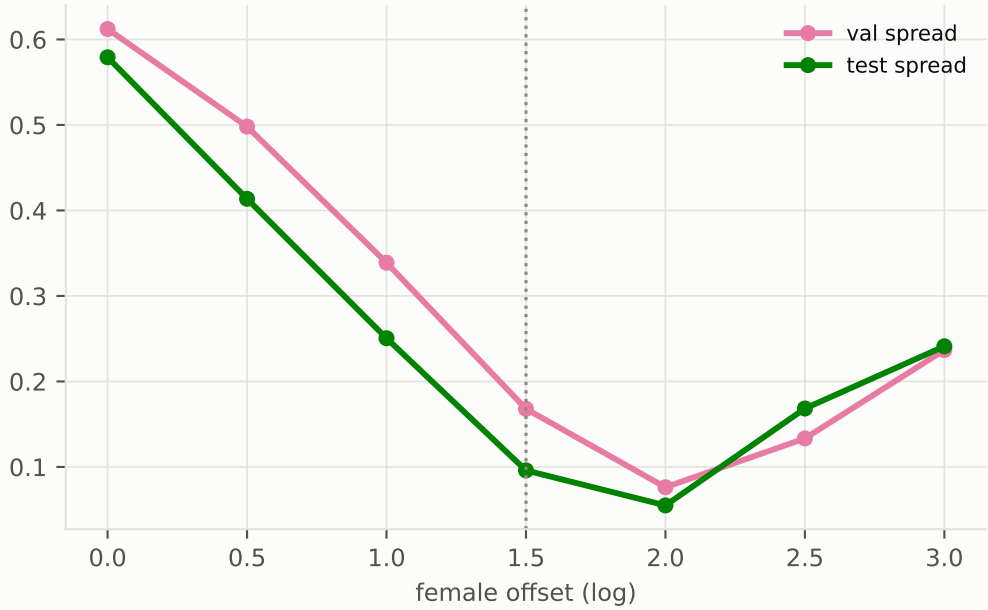
Calibration: offsets and confidence

Female log-offset sweep (child offset 1.0) and how trustworthy the score is, by confidence band.

Macro-F1 vs female offset

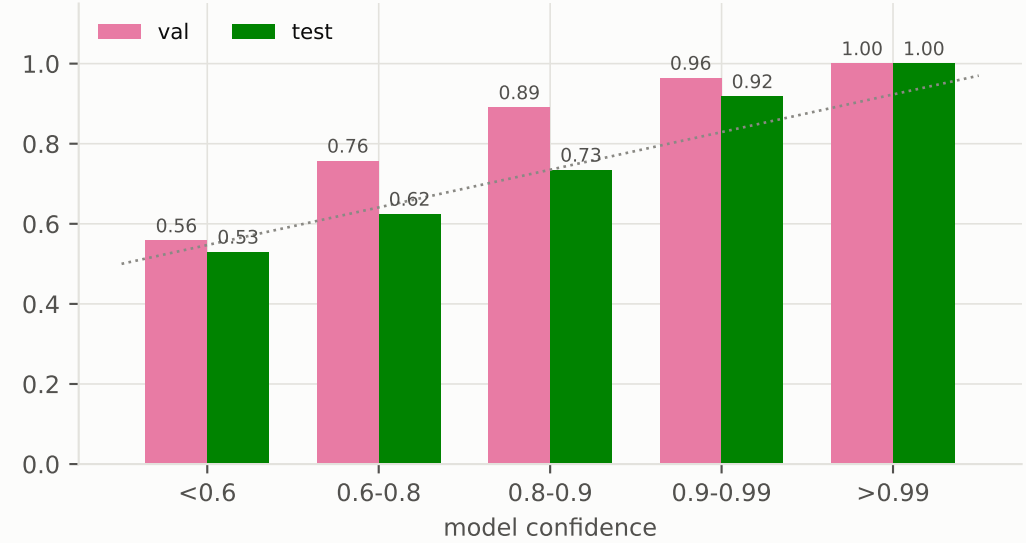


Recall spread vs female offset (lower = more even)

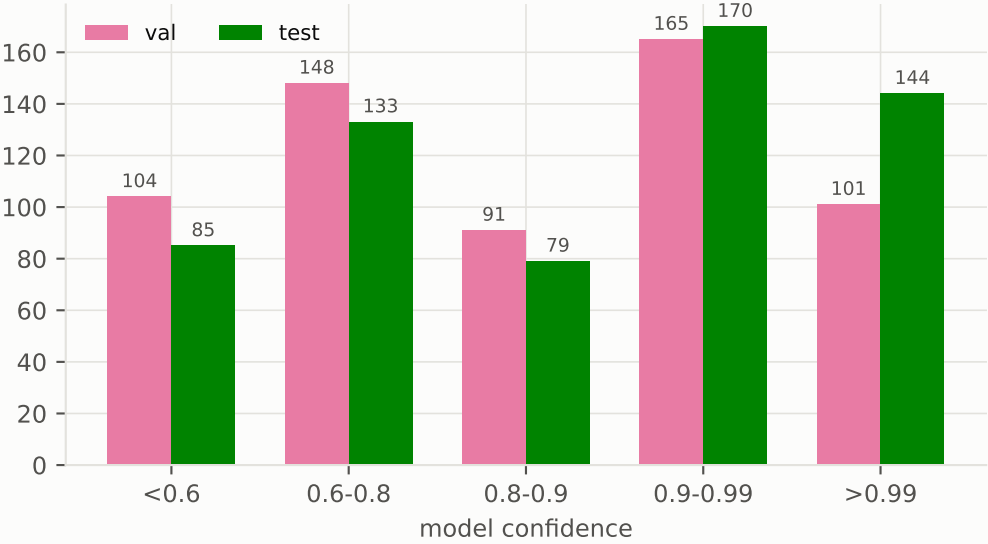


A score of 0.8-0.9 on test is right only about 3 times in 4; only scores above 0.99 are dependable. That, not the dashboard, explains the gap between shown scores and real accuracy.

Accuracy inside each confidence band



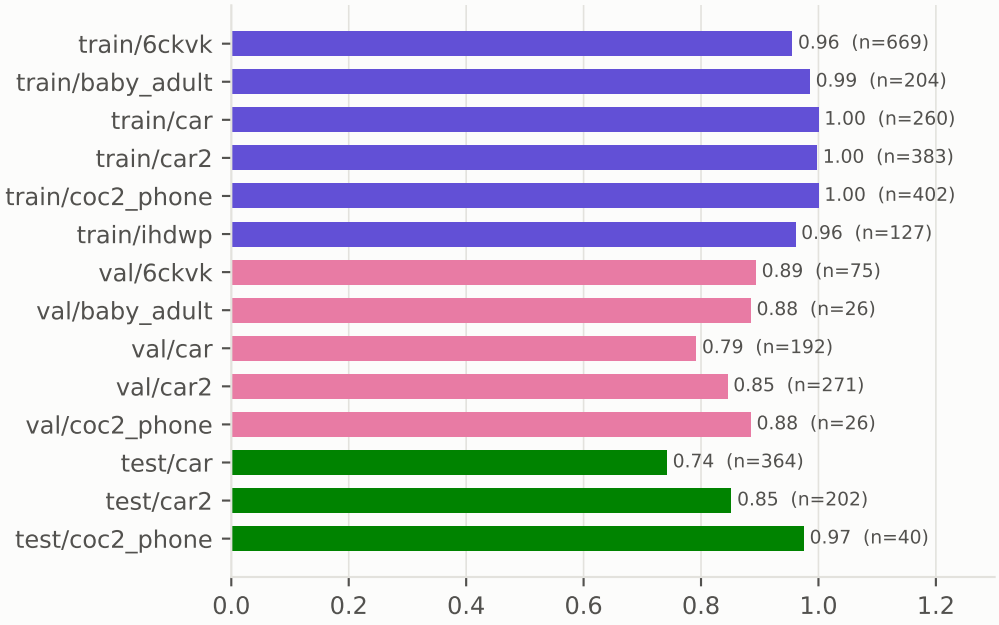
How many crops fall in each band



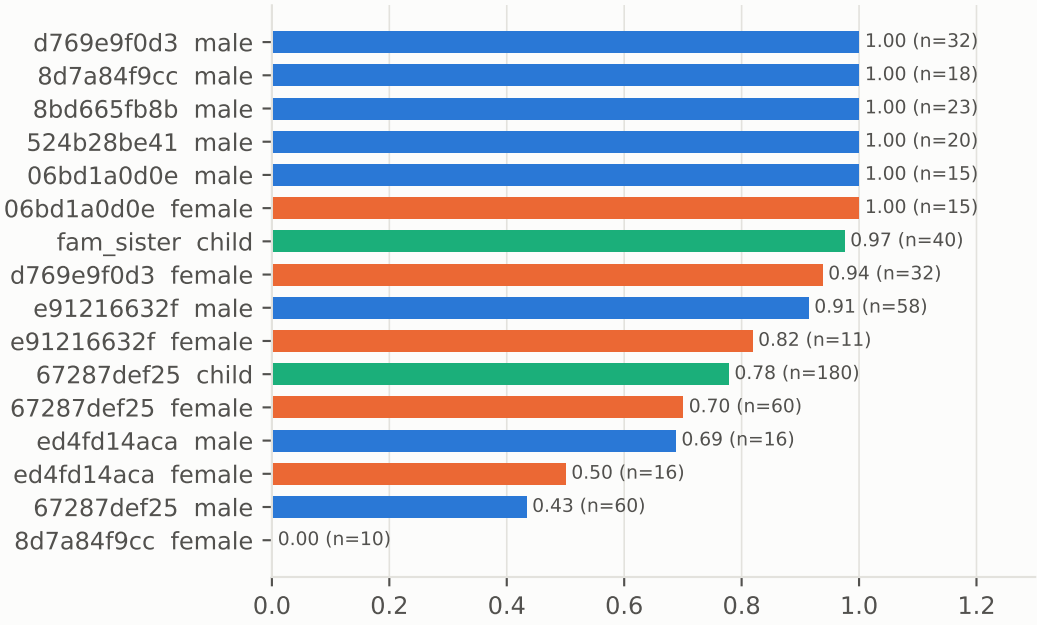
Where accuracy is lost: by data source, test group and live video

Same deployed model. Accuracy on training sources is near 100%; on unseen cabin footage it drops.

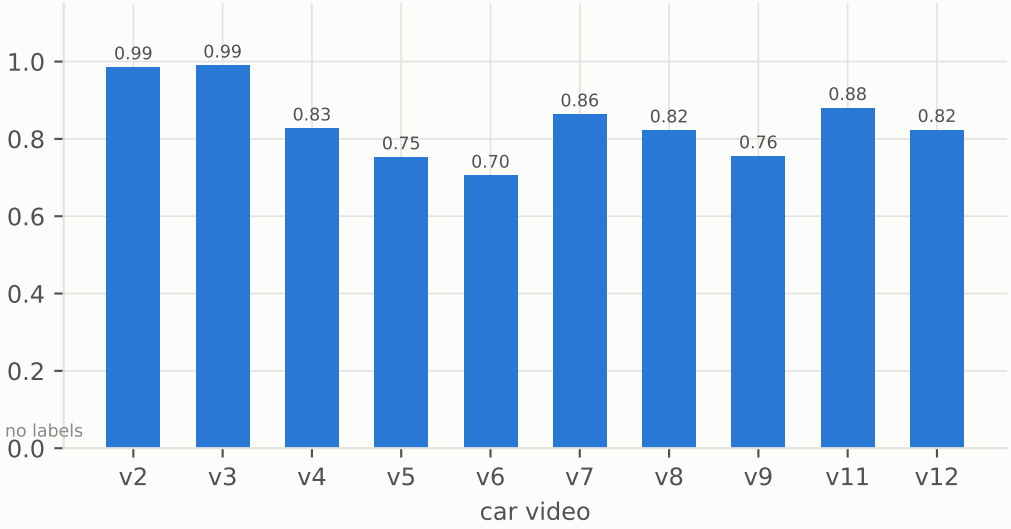
Accuracy by split / source



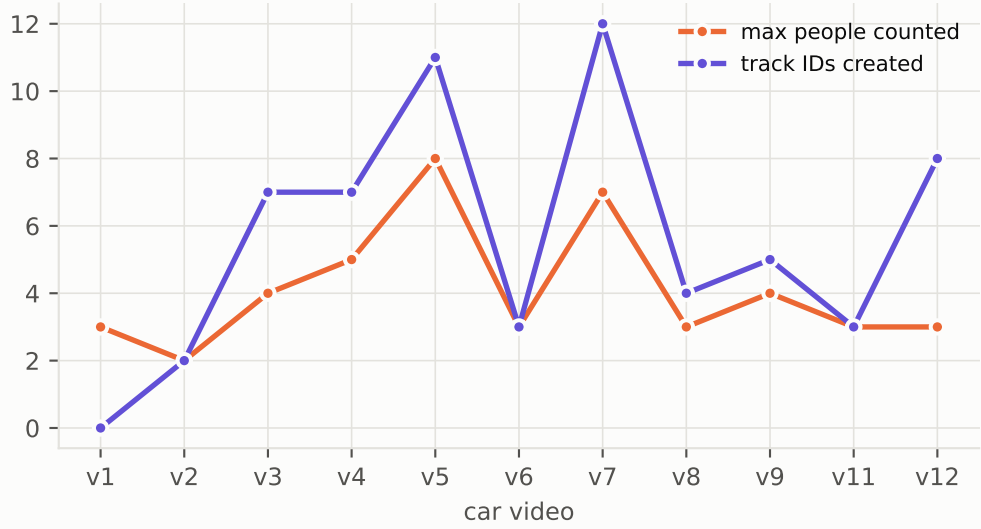
Test accuracy per group and class (n>=10)



Live run: frames agreeing with each person's main label



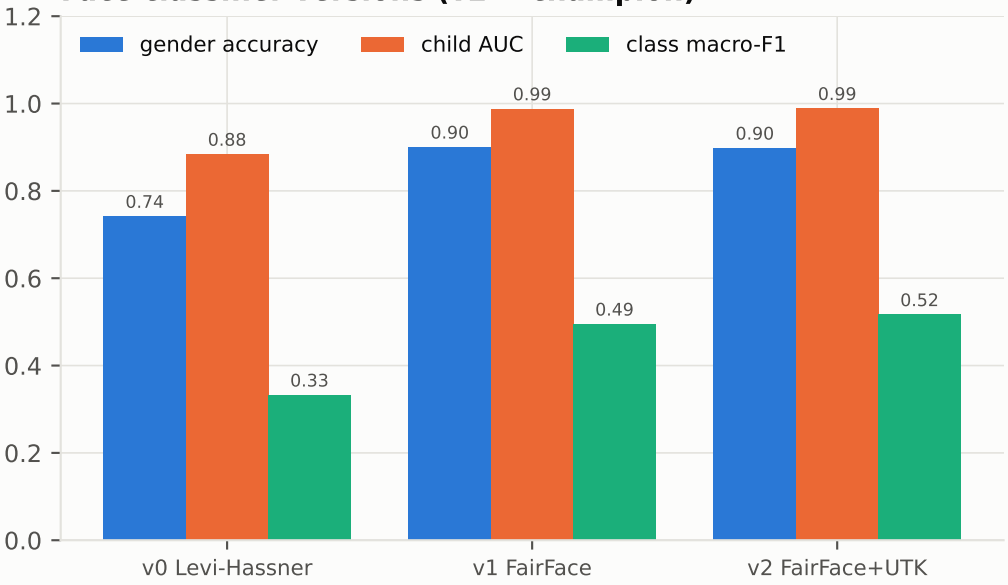
Live run: people counted vs track IDs created



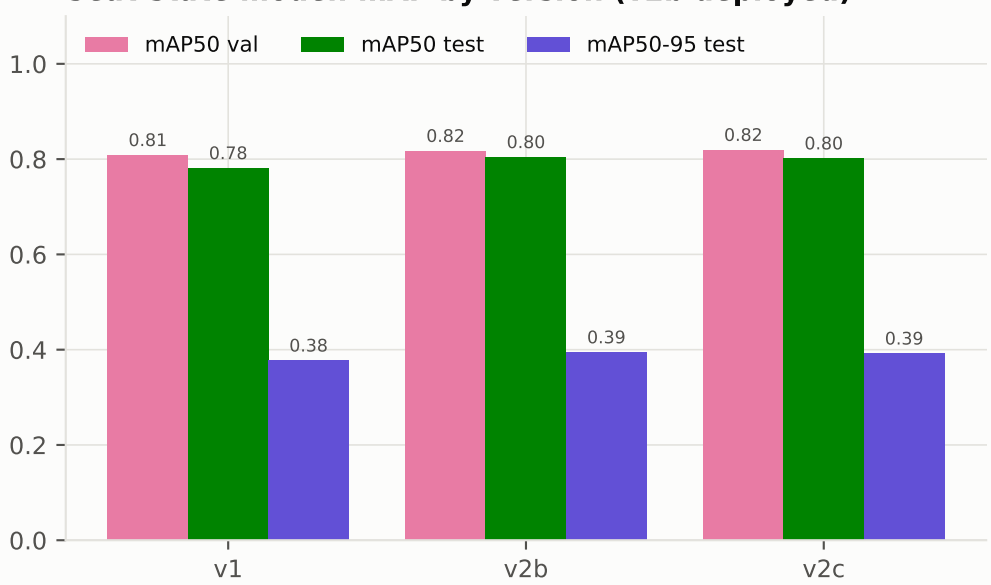
Face classifier and seat-state model (bus profile)

Registry and metrics files for the models that feed the bus profile.

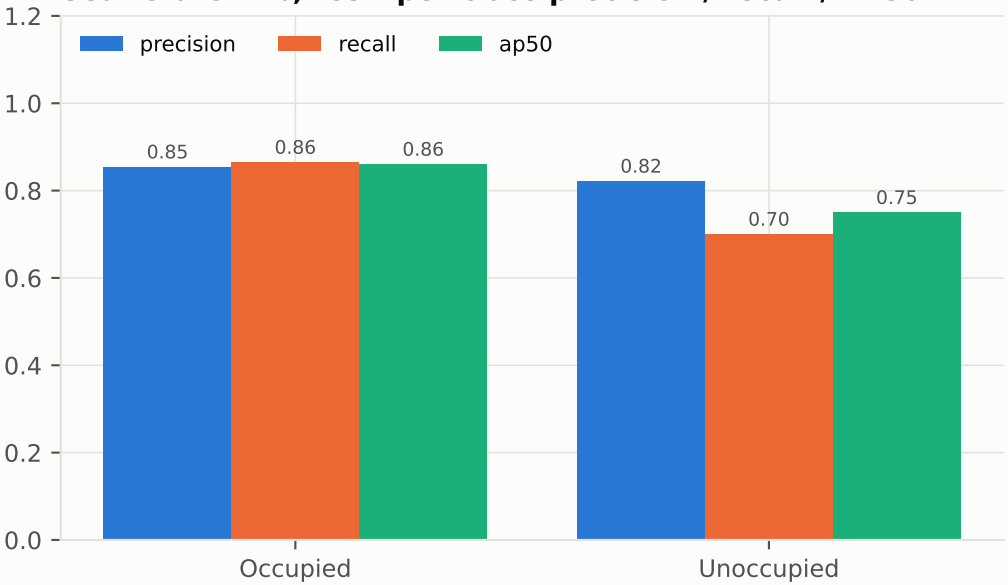
Face classifier versions (v2 = champion)



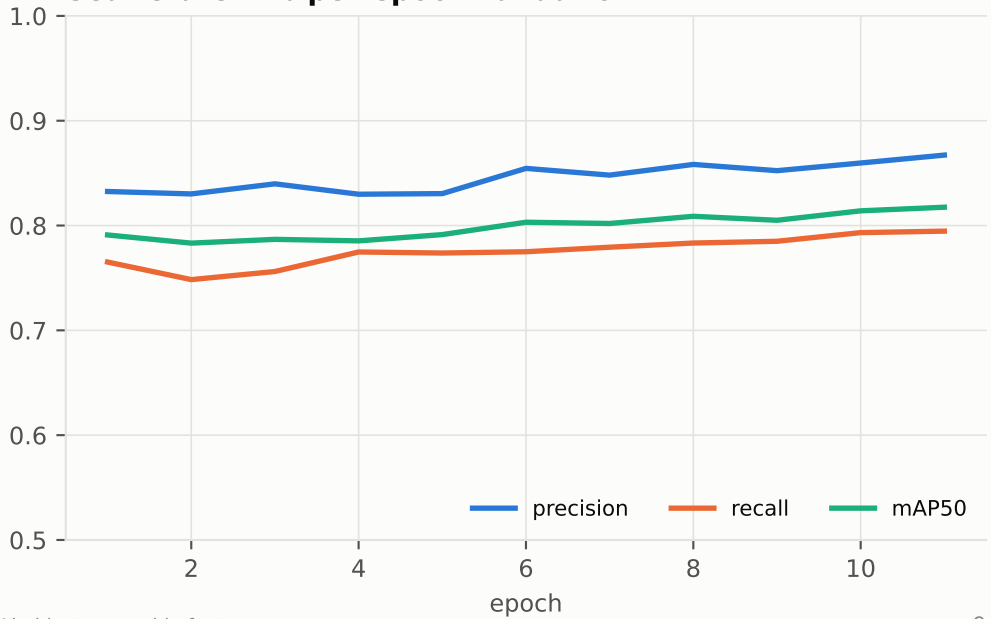
Seat-state model: mAP by version (v2b deployed)



Seat-state v2b, test: per-class precision / recall / AP50



Seat-state v2b per-epoch validation



Findings and what to do next

Plain-language reading of the charts in this report.

What the numbers say

The classifier fits its training data almost perfectly (98% accuracy) and reaches about 84% on validation and 80% on test. The gap, together with validation loss rising from about epoch 14 while train loss keeps falling, is overfitting to the small number of distinct people in training.

Calibration offsets lifted female recall from 0.53 to 0.72 on test and brought the recall spread down from 0.38 to about 0.10-0.16 (validation 0.03 at the earlier 2.0 / 0.5 setting). The cost is some male and child recall. No offset serves every class.

Model confidence is only trustworthy above 0.99. On test, the 0.8-0.9 band is right about 73% of the time and 0.6-0.8 only 62%, so the scores shown on a dashboard overstate accuracy for cabin footage.

The fine-tuned person detector raised recall but counts worse than plain COCO YOLOv8n, especially on crowded frames (5 people), because training never had more than 3 people in a frame.

In the live run on 11 car videos, simple clips are labelled consistently, while crowded clips show label flips (e.g. one person read as male, female and child) and more track IDs than people.

What would improve the model, in priority order

1. Record test and training footage from your own camera: 15-20 women, 15-20 men, 5+ children aged 2-10, front and rear seats, night/IR and crowded cabins. Keep a person-disjoint test set and look at it once.

2. Replace stock-photo training examples with real cabin examples, especially adult women and children. Balance by distinct people, not by crop count.

3. Label the 178 review crops and spot-check the 262 face-labelled training crops; get a second reviewer for a sample of the cabin labels.

4. Fix the crop preprocessing (neutral padding instead of edge replication) and add cabin-like augmentation: blur, low light, IR, partial occlusion. Stop training near epoch 14 and freeze the backbone early.

5. Add crowded frames (4-6 people) to detector data; re-measure count error and ID switches against hand-counted frames.

6. Re-tune offsets and an 'unknown' threshold on a validation set from your camera, then measure once on the held-out test. Measure latency on the Raspberry Pi before enabling.

Caveats

Test is about 20 people (6 women, 8 men and 7-8 children groups), so one person moves a class recall by 10-30 points. The test set has been viewed several times, so tuning on it is no longer independent.

Per-epoch classifier accuracy was not logged during training; validation loss and macro-F1 are shown instead. Metrics for v3 and v4 seeds in the history charts are validation only. Train/val/test figures for the deployed model are recomputed here from the saved crops and weights.